

# CSA0404 – OPERATING SYSTEMS – LAB PROGRAMS

## EXPERIMENT 19

**Question:** Design a C program to implement process synchronization using mutex locks.

**Program:**

```
1 #include <stdio.h>
2 #include <pthread.h>
3
4 int counter = 0;
5 pthread_mutex_t mutex;
6
7 void *threadFunction(void *arg) {
8     int i;
9     for (i = 0; i < 1000000; ++i) {
10         pthread_mutex_lock(&mutex);
11         counter++;
12         pthread_mutex_unlock(&mutex);
13     }
14     return NULL;
15 }
16
17 int main() {
18     pthread_mutex_init(&mutex, NULL);
19
20     pthread_t thread1, thread2;
21
22     pthread_create(&thread1, NULL, threadFunction, NULL);
23     pthread_create(&thread2, NULL, threadFunction, NULL);
24
25     pthread_join(thread1, NULL);
26     pthread_join(thread2, NULL);
27
28     pthread_mutex_destroy(&mutex);
29
30     printf("Final counter value: %d\n", counter);
31     return 0;
32 }
```

Final counter value: 2000000

...Program finished with exit code 0  
Press ENTER to exit console.